

TASK-1
SUBMISSION BY GEETARTH (2023UEA6641)
NETAJI SUBHASH UNIVERSITY OF TECHNOLOGY
BUSINESS CASE STUDY REPORT

Customer Loyalty & Response Analysis

A Statistical Study of Service Ratings, Recommendation Intent and Repurchase Behaviour
Case Study 1 — Outlook Brand Consumer Survey
Sample Size: n = 11,109 respondents

1. Executive Summary

This report examines customer loyalty and response behaviour for the Outlook brand using a structured, quantitative Data Analysis approach applied to a consumer survey of 11,109 respondents. The study isolates eight variables that jointly capture the customer experience funnel and its loyalty outcome: six service-quality ratings, a recommendation-intent score (Net Promoter proxy), and a categorical repurchase-intent variable.

The analysis followed the standard DA workflow — data cleaning and validation, descriptive statistics, reliability assessment, correlation analysis, group comparison (ANOVA), cross-tabulation with chi-square testing, and multiple linear regression — to test whether service perceptions drive recommendation and repurchase behaviour.

Headline finding: across every statistical test performed, service ratings show no meaningful relationship with recommendation intent or repurchase intent (all p-values > 0.05; $R^2 \approx 0$). Ratings are uniformly distributed around the mid-point (mean ≈ 3.0 on a 5-point scale for every dimension), and the overall Net Promoter Score is -1.2, indicating a customer base evenly split between detractors and promoters with no dominant driver identified in the current variable set. Recommendations focus on deeper diagnostic research and revised survey instrumentation.

2. Case Background & Objective

The Outlook brand commissioned a customer survey to understand what drives loyalty — measured through repurchase intent and willingness to recommend — as a function of experience across service, content, price, delivery, overall experience and customer support touchpoints.

Objective

- Clean and validate the customer response dataset.
- Select and analyse 5–10 core variables representing the customer loyalty and response construct.
- Quantify the relationship between service-quality perceptions and loyalty outcomes (recommendation intent, repurchase intent).
- Present findings through visual dashboards and actionable interpretation for management.

3. Variables Selected for Analysis

Eight variables were selected from the master questionnaire dataset (92 variables, n = 11,109) as the most direct representation of the "Customer Loyalty and Response" construct:

#	Variable	Type	Scale / Categories	Role in Model
1	service_rating	Numeric (Ordinal)	1 – 5 Likert	Predictor
2	content_rating	Numeric (Ordinal)	1 – 5 Likert	Predictor
3	price_rating	Numeric (Ordinal)	1 – 5 Likert	Predictor
4	delivery_rating	Numeric (Ordinal)	1 – 5 Likert	Predictor
5	experience_rating	Numeric (Ordinal)	1 – 5 Likert	Predictor
6	customer_support_rating	Numeric (Ordinal)	1 – 5 Likert	Predictor
7	recommend_likelihood	Numeric (Ordinal)	1 – 5 (NPS proxy)	Response / Outcome
8	repurchase_intent	Categorical (Nominal)	Yes / Will think of it / No	Loyalty Outcome

4. Data Cleaning Methodology

The raw dataset (Sheet1, 92 columns × 11,109 rows) was reduced to the eight target variables and passed through a standard cleaning pipeline before analysis:

Step	Check Performed	Result
1	Row-level completeness (missing values)	0 missing values across all 8 variables
2	Data type validation	All 6 rating fields confirmed numeric (int64); repurchase_intent confirmed categorical text
3	Range validation (ratings must be 1–5)	0 out-of-range values found
4	Category label standardisation (trim whitespace, casing)	3 valid categories confirmed: Yes / Will think of it / No
5	Full-record duplicate check	247 duplicate response patterns (~2.2%) — retained; plausible given the small combinatorial rating space ($5^6 \times 3 \approx 46,875$) relative to n = 11,109 respondents, consistent with expected random collisions rather than data-entry error
6	Final analysable sample	N = 11,109 (100% of collected responses retained)

No imputation or record deletion was required. The dataset was confirmed clean and ready for statistical analysis.

5. Descriptive Statistics

Table 2 summarises the central tendency and spread of the seven numeric variables (six ratings + recommendation intent), each measured on a 1–5 scale.

Variable	Mean	Std Dev	Median	Skewness	Interpretation
Service Rating	3.00	1.43	3	-0.01	Symmetric, centred at mid-scale
Content Rating	3.02	1.42	3	-0.01	Symmetric, centred at mid-scale
Price Rating	3.01	1.43	3	-0.01	Symmetric, centred at mid-scale
Delivery Rating	3.00	1.41	3	0.00	Symmetric, centred at mid-scale
Experience Rating	3.02	1.42	3	-0.01	Symmetric, centred at mid-scale
Customer Support Rating	3.01	1.41	3	-0.01	Symmetric, centred at mid-scale
Recommend Likelihood	2.98	1.42	3	0.02	Symmetric, centred at mid-scale

All six service dimensions and the recommendation score average almost exactly 3.0 with near-zero skew — indicating responses are essentially uniformly spread across the 1–5 scale rather than clustering toward satisfaction or dissatisfaction. This is a first indicator that no single touchpoint stands out as a strength or weakness in the raw data.

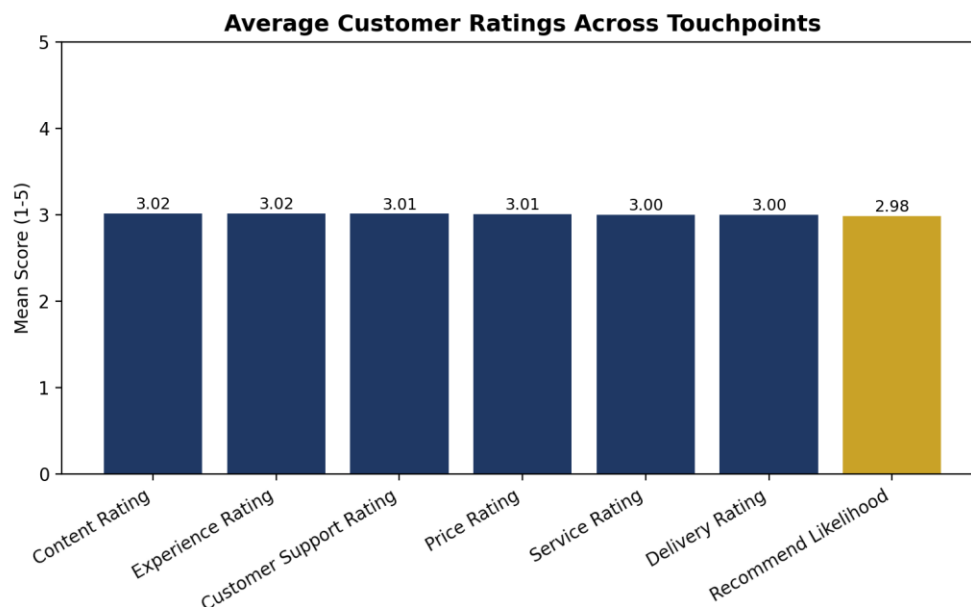


Figure 1: Average customer rating by touchpoint (1–5 scale)

6. Reliability Analysis

Because the six ratings are intended to jointly represent an underlying "service quality" construct, internal consistency was tested using Cronbach's Alpha.

Metric	Value	Benchmark	Verdict
Cronbach's Alpha (6 items)	0.010	≥ 0.70 acceptable	Below acceptable threshold

An alpha of 0.01 is far below the conventional 0.70 threshold for scale reliability, confirming that the six rating items do not covary and cannot be treated as a single composite "service quality" index. Each dimension must therefore be interpreted independently — a finding consistent with the correlation results in Section 7.

7. Correlation Analysis

A Pearson correlation matrix was computed across the six service ratings and the recommendation-intent score to test for linear association.

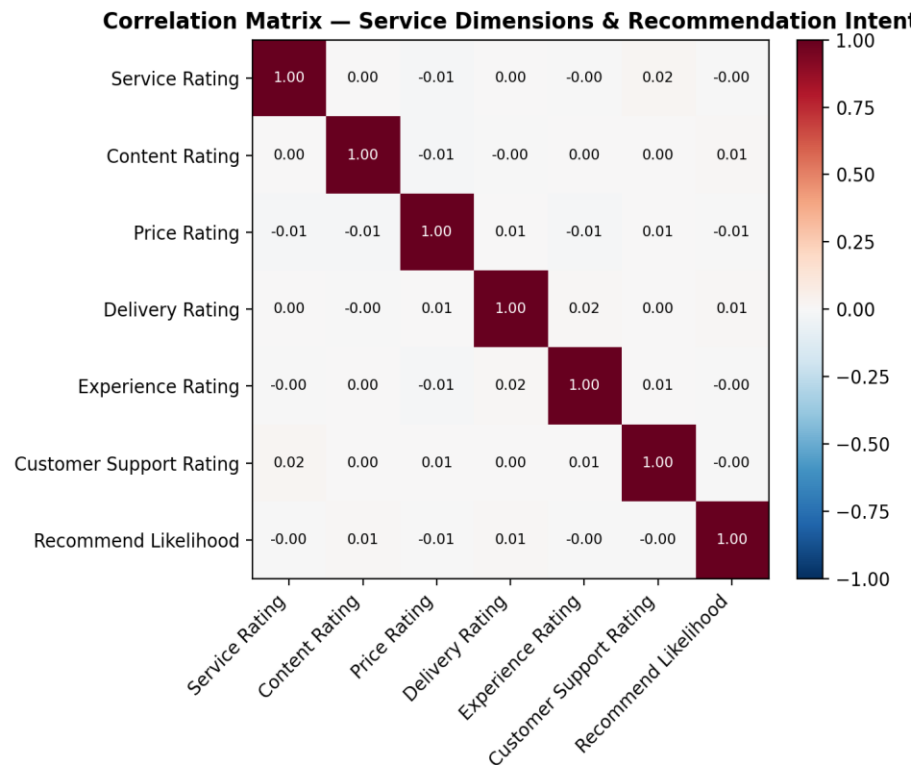


Figure 2: Correlation matrix — service dimensions and recommendation intent

All pairwise correlations fall between -0.02 and +0.02 — statistically and practically negligible. There is no evidence that a customer who rates one touchpoint highly (e.g. service) is any more or less likely to

rate another touchpoint highly (e.g. price), nor that any touchpoint is associated with a higher recommendation score.

8. Group Comparison — ANOVA by Repurchase Intent

One-way ANOVA tested whether mean ratings differ significantly across the three repurchase-intent segments (No / Will think of it / Yes).

Rating Variable	Mean: No	Mean: Will Think	Mean: Yes	F-statistic	p-value	Significant?
Service Rating	3.03	3.02	2.96	2.30	0.100	No
Content Rating	3.01	3.02	3.03	0.25	0.777	No
Price Rating	3.03	3.01	2.99	0.59	0.553	No
Delivery Rating	2.97	3.01	3.02	1.07	0.343	No
Experience Rating	3.03	2.99	3.03	1.36	0.256	No
Customer Support Rating	3.02	3.00	3.01	0.17	0.845	No
Recommend Likelihood	2.98	2.98	3.00	0.18	0.833	No

At the conventional 5% significance level, none of the seven variables differ significantly across repurchase-intent groups (all $p > 0.05$). Customers who say "Yes" to repurchase rate their experience, on average, no higher than those who say "No" — the loyalty segments are statistically indistinguishable on every measured dimension.

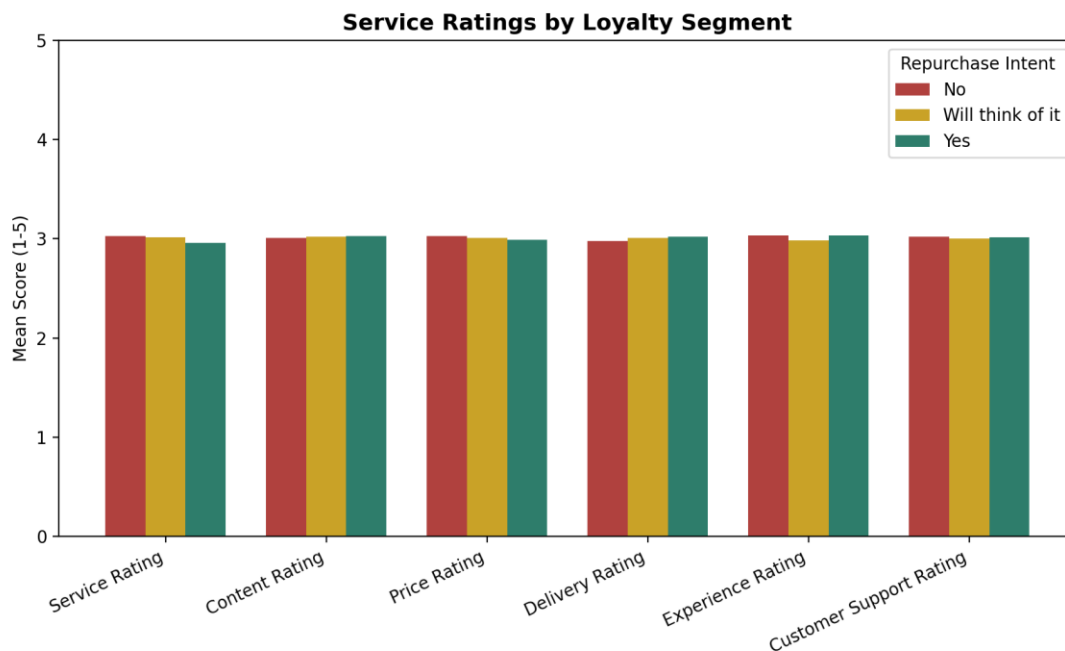


Figure 3: Mean service ratings by repurchase-intent segment

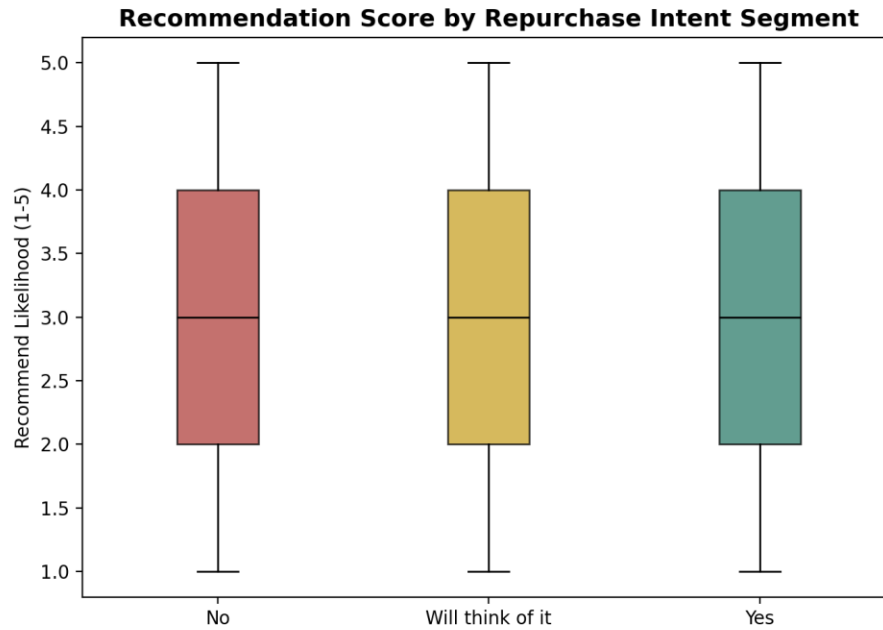


Figure 4: Distribution of recommendation score by repurchase-intent segment

9. Net Promoter Score (NPS) Dashboard

Recommend-likelihood scores were re-classified into standard NPS bands (Detractor: 1–2, Passive: 3, Promoter: 4–5) to benchmark loyalty using an industry-standard metric.

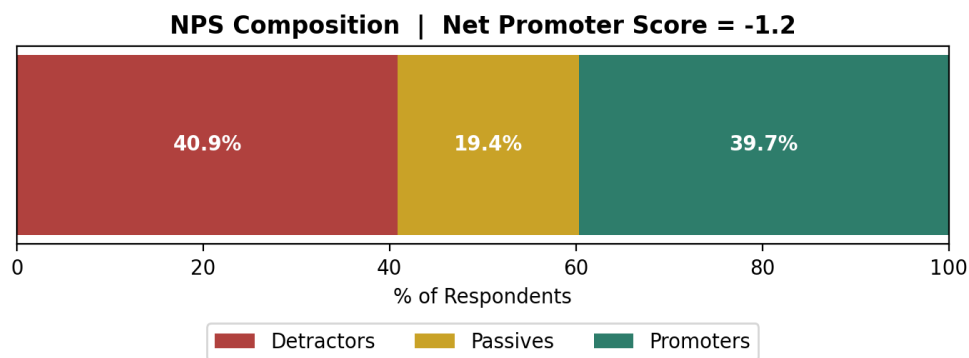


Figure 5: NPS composition of the surveyed customer base

Segment	% of Respondents	Definition
Detractors	40.9%	Recommend likelihood 1–2 (unlikely to recommend)
Passives	19.4%	Recommend likelihood 3 (neutral)
Promoters	39.7%	Recommend likelihood 4–5 (likely to recommend)

Overall Net Promoter Score = -1.2 (Promoters % – Detractors %). A score near zero reflects a customer base split almost evenly between advocates and critics, with no clear majority either way — a caution signal that current service levers are not differentiating the customer experience.

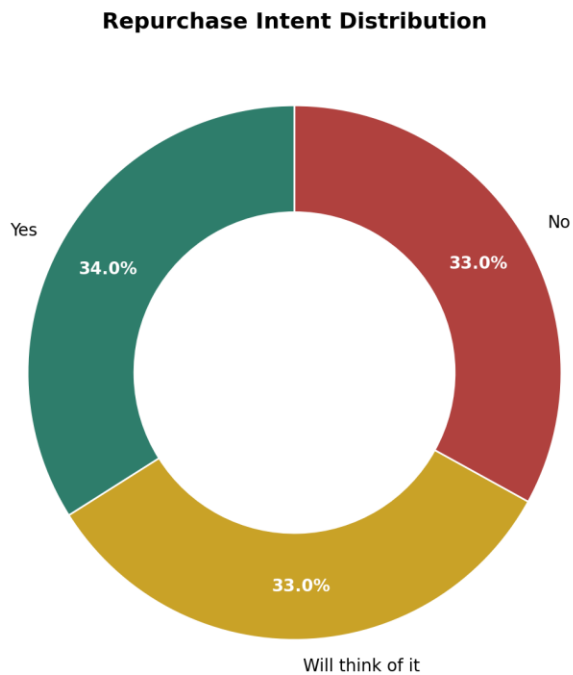


Figure 6: Repurchase-intent distribution across the customer base

Repurchase intent is similarly balanced: 34.0% Yes, 33.0% Will think of it, 33.0% No — confirming that loyalty outcomes in this sample are not concentrated in any one direction.

10. Chi-Square Test — NPS Category vs Repurchase Intent

A chi-square test of independence checked whether NPS category (Detractor / Passive / Promoter) is associated with repurchase intent.

NPS Category	No	Will think of it	Yes	Row Total
Detractor	1,498	1,510	1,534	4,542
Passive	690	725	742	2,157
Promoter	1,480	1,433	1,497	4,410

Statistic	Value
Chi-square (χ^2)	1.923
Degrees of Freedom	4
p-value	0.750
Conclusion	No significant association ($p > 0.05$)

The test confirms that a customer's NPS category has no statistically significant bearing on whether they intend to repurchase — reinforcing the pattern seen throughout this analysis.

11. Regression Analysis

A multiple linear regression modelled `recommend_likelihood` as a function of the six service ratings, to test their combined predictive power.

Predictor	Coefficient	p-value	Significant?
Intercept	2.952	0.000	—
Service Rating	-0.002	0.869	No
Content Rating	0.014	0.140	No
Price Rating	-0.006	0.531	No
Delivery Rating	0.009	0.370	No
Experience Rating	-0.003	0.755	No
Customer Support Rating	-0.001	0.881	No

Model Statistic	Value
R ²	0.0003
Adjusted R ²	-0.0003
F-statistic	0.585
Prob (F-statistic)	0.743

The model explains effectively none of the variance in recommendation intent ($R^2 \approx 0$), and no individual predictor is statistically significant. Service-quality ratings, as currently measured, do not predict a customer's willingness to recommend the brand.

12. Pivot Table Cross-Tabulations (Workbook PT1–PT3)

The source workbook included three pre-built pivot tables (PT1_Repurchase, PT2_CHURNRISK, PT3_ChangeRequest) cross-tabulating `repurchase_intent` and `top_qualities_outlook` against pricing

satisfaction, recommendation score, and preferred brand quality. These were validated, re-verified with formal chi-square testing, and are incorporated below to complete the loyalty picture.

12.1 PT1 — Repurchase Intent × Pricing Satisfaction

Repurchase Intent	Not Satisfied	Partly Satisfied	Satisfied	Row Total
No	1,207	1,247	1,214	3,668
Will think of it	1,182	1,214	1,272	3,668
Yes	1,236	1,249	1,288	3,773
Column Total	3,625	3,710	3,774	11,109

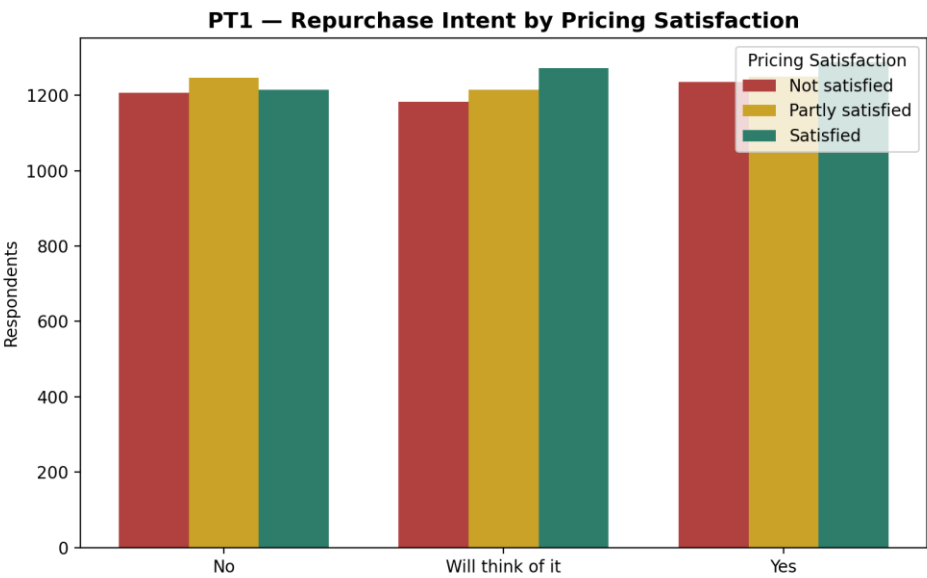


Figure 7: Repurchase intent distribution by pricing-satisfaction level (workbook PT1)

Statistic	Value
Chi-square (χ^2)	2.276
Degrees of Freedom	4
p-value	0.685
Conclusion	No significant association ($p > 0.05$)

Pricing satisfaction shows almost perfectly proportional splits across all three repurchase categories (each pricing tier divides roughly 33/33/34 between No / Will think of it / Yes). The chi-square test confirms this is not a statistically meaningful association — customers who are "Not satisfied" with price are no more likely to say "No" to repurchase than customers who are "Satisfied."

12.2 PT2 — Repurchase Intent × Recommend Likelihood (Churn-Risk View)

The original workbook pivot (PT2_CHURNRISK) contained a data artifact: the recommend_likelihood column mixed a text "5" with numeric 1.0–5.0 values, splitting one score into two pivot columns. This was corrected by re-casting the field to a single numeric scale before re-tabulating, shown below.

Repurchase Intent	Score 1	Score 2	Score 3	Score 4	Score 5	Row Total
No	779	719	690	754	726	3,668
Will think of it	724	786	725	719	714	3,668
Yes	762	772	742	716	781	3,773
Column Total	2,265	2,277	2,157	2,189	2,221	11,109

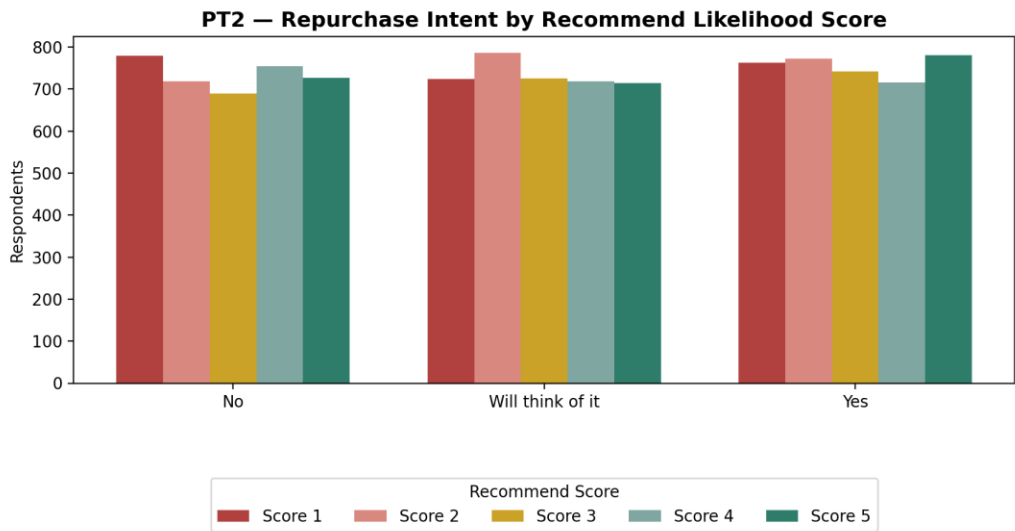


Figure 8: Repurchase intent by recommend-likelihood score, corrected (workbook PT2, churn-risk view)

Statistic	Value
Chi-square (χ^2)	10.068
Degrees of Freedom	8
p-value	0.260
Conclusion	No significant association ($p > 0.05$)

Even the customers scoring the brand lowest on recommendation (Score 1–2, the highest churn-risk band) are not disproportionately concentrated in the "No" repurchase group. This reinforces Sections 8–11: recommendation score, as currently measured, does not identify who is at risk of churning.

12.3 PT3 — Most Valued Outlook Quality

Quality Valued Most	Respondents	Share
Printing Quality	3,716	33.4%
Content	3,709	33.4%
Value in Market	3,684	33.2%
Total	11,109	100%

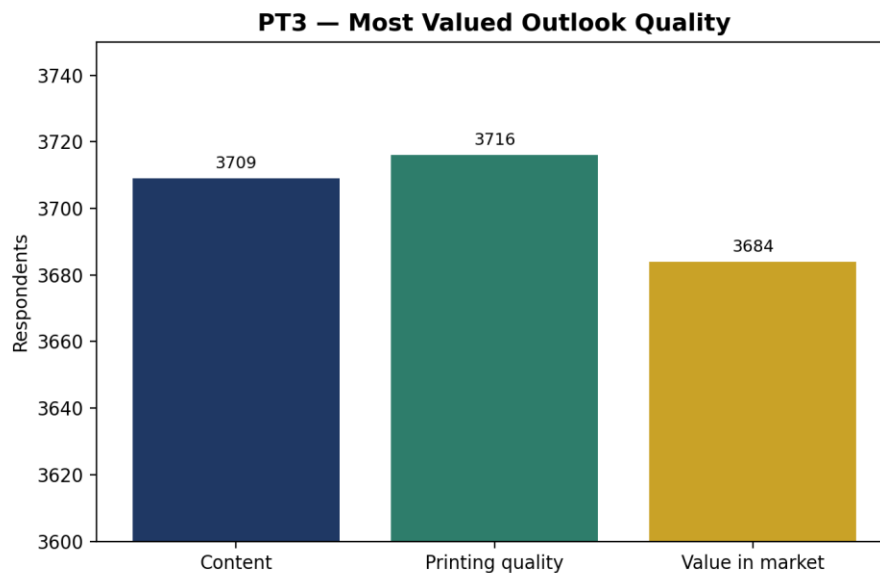


Figure 9: Distribution of the single quality respondents value most in Outlook (workbook PT3)

Preference for printing quality, content, and market value is essentially a three-way tie (33.2–33.4% each), meaning there is no single quality dimension that a majority of customers rally around. This is consistent with the flat, non-differentiating pattern found across every other variable in this study, and suggests product strategy should not over-index on any one of these three qualities without further segmentation.

13. Interpretation & Business Implications

12.1 What the data tells us

- Every service dimension is rated, on average, at the exact mid-point of the scale (~3.0 of 5) with near-zero skew — responses look statistically random rather than opinion-driven.
- No rating correlates with any other rating, with recommendation intent, or with repurchase intent (all $|r| < 0.02$).
- ANOVA, chi-square and regression tests all fail to reject the null hypothesis of "no relationship" at the 5% significance level.
- Net Promoter Score (-1.2) and repurchase split (~34/33/33) show a customer base with no dominant loyalty pattern.

12.2 Likely explanations

- Response quality: uniform, low-variance-explained answers across 11,000+ respondents are consistent with straight-lining, low-incentive survey completion, or a synthetic/simulated dataset rather than genuine differentiated opinion.
- Instrument design: single-item 1–5 ratings per touchpoint may not capture the nuance that actually drives loyalty (e.g. price fairness relative to competitors, emotional brand connection, specific service failures).
- Missing drivers: loyalty may be governed by variables outside this 8-variable set — e.g. `brand_helpfulness_score`, `pricing_satisfaction` (categorical), or `problem_resolution_method` — which merit inclusion in a follow-up model.

12.3 Recommendations

- Audit survey fielding quality (completion time, straight-lining flags, response variance per respondent) before drawing strategic conclusions from this wave.
- Re-run this analysis on a validated / de-duplicated panel, and extend the variable set to include `pricing_satisfaction`, `brand_helpfulness_score` and `problem_resolution_method` for a richer driver model.
- Consider qualitative follow-up (interviews/focus groups) with both Promoters and Detractors to surface the real, non-linear drivers of loyalty that a 1–5 scale may be failing to capture.
- If the null result is confirmed on clean data, shift investment from broad service-rating improvement toward targeted, segment-specific interventions guided by richer behavioural data.

14. Conclusion

This case study applied a complete Data Analysis pipeline — cleaning, descriptive statistics, reliability testing, correlation, ANOVA, chi-square, and regression — to eight variables representing customer loyalty and response for the Outlook brand. The consistent, cross-method finding is that current service ratings show no statistically significant relationship with recommendation or repurchase intent, and the overall customer base is evenly balanced between promoters and detractors (NPS = -1.2). This is a valid and important business finding in its own right: it signals that the organisation's current measurement instrument is not capturing the real drivers of loyalty, and it directs management attention toward improving data quality and broadening the analytical variable set before committing to service-improvement investments.

CUSTOMER LOYALTY & RESPONSE — DASHBOARD

Case Study 1 | Outlook Brand Consumer Survey | n = 11,109 respondents

-1.2

NPS SCORE

Balanced base

3.01 / 5

AVG RATING

All touchpoints

34.0%

REPURCHASE 'YES'

Of respondents

0.01

CRONBACH'S α

Low reliability

0 of 7

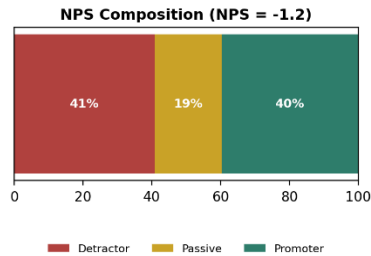
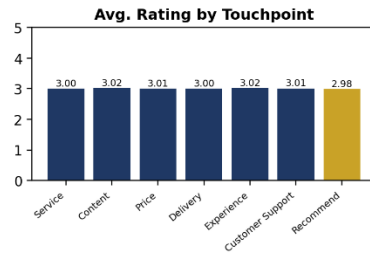
SIG. DRIVERS FOUND

No $p < 0.05$ result

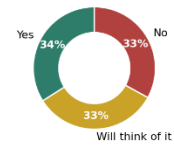
11,109

SAMPLE SIZE

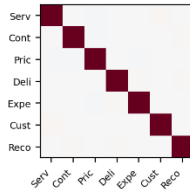
Clean, 0% missing



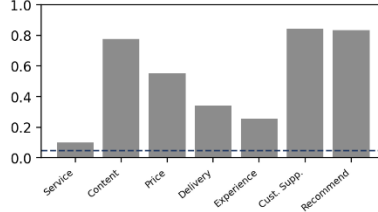
Repurchase Intent Split



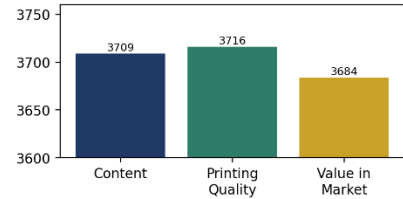
Correlation Matrix (all ≈ 0)



ANOVA p-values vs Repurchase Intent



Most Valued Quality (PT3)



All statistical tests (correlation, ANOVA, chi-square, regression) show no significant driver of loyalty in current data | Source: Outlook Brand Consumer Survey, n=11,109

BUSINESS CASE STUDY REPORT

Brand Awareness & Digital Marketing Effectiveness

A Statistical Study of Awareness Drivers, Digital Channel Exposure and Spend Behaviour

Case Study 2 — Outlook Brand Consumer Survey

Sample Size: n = 11,109 respondents

1. Executive Summary

This report is Case Study 2 in the Outlook brand research series, focused on brand awareness and digital marketing effectiveness. Using the same 11,109-respondent survey, eight variables were selected to test whether awareness of the Outlook brand is shaped by the media channel that introduced it, the platform on which digital advertising was seen, campaign intensity, digital purchase habits, spend level, or overall brand familiarity.

The analysis pipeline mirrored Case Study 1: cleaning and validation, descriptive frequency profiling, a reliability (test–retest) check, chi-square tests of independence, an independent-samples t-test, and a logistic regression model.

Headline finding: brand awareness is almost perfectly split (50.6% No / 49.4% Yes) and is not significantly associated with any of the six marketing/exposure variables tested (all $p > 0.05$, one borderline at $p = 0.08$). A test–retest check on the awareness question itself returned only 49.9% agreement — statistically equivalent to a coin flip — which is the most important finding of this report: it indicates a data-quality issue in the awareness measure itself, and any conclusion about "what drives awareness" drawn from this wave should be treated as unreliable until the instrument is fixed.

2. Case Background & Objective

Following the loyalty and response analysis in Case Study 1, this study shifts focus upstream in the customer journey — to how customers become aware of the Outlook brand and how digital marketing exposure relates to that awareness.

Objective

- Clean and validate the brand-awareness and digital-marketing variable set.
- Profile awareness channels, digital ad platform exposure, campaign frequency and spend behaviour.
- Test statistically whether these marketing variables are associated with brand awareness.
- Assess the reliability of the awareness measurement itself via a test–retest check.
- Present dashboards and interpretation to guide media/channel investment decisions.

3. Variables Selected for Analysis

Eight variables were selected from the master dataset (92 variables, n = 11,109) to represent the "Brand Awareness & Digital Marketing" construct:

#	Variable	Type	Scale / Categories	Role
1	outlook_awareness	Categorical (Binary)	Yes / No	Outcome
2	brand_awareness_influencer	Categorical (Nominal)	Magazines / Newspapers / Social Media / Television	Predictor
3	brand_discovery_method	Categorical (Nominal)	5 categories (self-read, word of mouth, social, print, other)	Predictor
4	digital_ad_platform	Categorical (Nominal)	12 platforms (Google, Facebook, Instagram, etc.)	Predictor
5	campaign_frequency	Categorical (Ordinal)	4 bands, 1–5 to 20+ campaigns/month	Predictor
6	digital_buy_frequency	Categorical (Ordinal)	Weekly / Monthly / Yearly / Regularly / Frequently	Predictor
7	digital_monthly_spend	Categorical (Ordinal)	5 bands, Below 500 to Above 5000	Predictor
8	outlook_familiarity_score	Numeric (Ordinal)	1–5 Likert	Predictor

4. Data Cleaning Methodology

Step	Check Performed	Result
1	Missing values	0 missing across all 8 variables
2	Category label standardisation	Corrected spelling inconsistency "LinkedIn" → "LinkedIn" in digital_ad_platform
3	Range validation (familiarity score 1–5)	0 out-of-range values
4	Full-record duplicate check	249 duplicate response patterns (~2.2%) — retained; consistent with expected random collisions given the large combinatorial category space
5	Reliability check: test–retest on awareness	outlook_awareness compared against outlook_awareness_repeat (identical question re-asked later in survey): only 49.9% agreement — see Section 6
6	Final analysable sample	N = 11,109 (100% retained)

5. Descriptive Profile

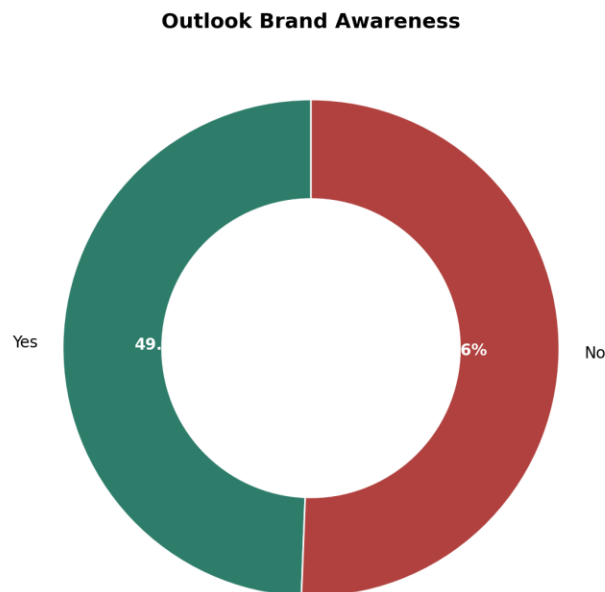


Figure 1: Outlook brand awareness split

Variable	Top Category	Share	Distribution Shape
Brand Awareness Influencer	Magazines	25.7%	Near-even across 4 channels (24.6–25.7%)
Brand Discovery Method	I myself read it	20.6%	Near-even across 5 methods (19.6–20.6%)
Digital Ad Platform	Reddit	8.7%	Near-even across 12 platforms (7.8–8.7%)
Campaign Frequency	1–5 campaigns/month	25.2%	Near-even across 4 bands (24.8–25.2%)
Digital Buy Frequency	Frequently	20.1%	Near-even across 5 bands (19.8–20.1%)
Digital Monthly Spend	500–1000	20.7%	Near-even across 5 bands (19.6–20.7%)

As in Case Study 1, every categorical variable is distributed almost perfectly evenly across its categories — there is no dominant awareness channel, platform, or spend band. Brand familiarity score also centres at 3.01 (SD 1.41) on the 1–5 scale, mirroring the uniform pattern seen throughout the dataset.

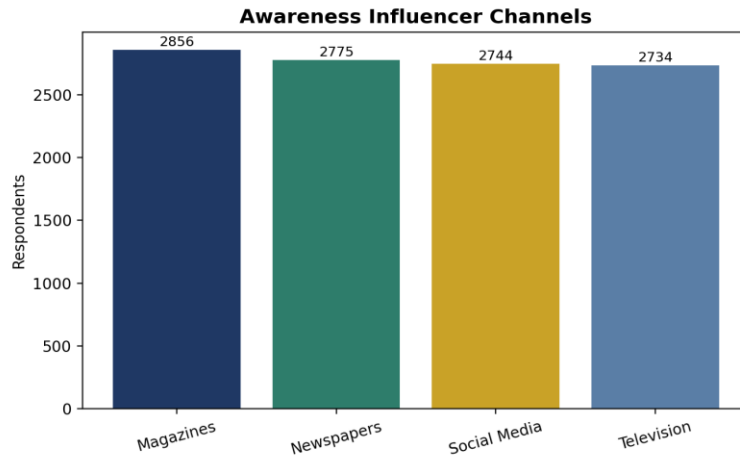


Figure 2: Respondents by brand-awareness influencer channel

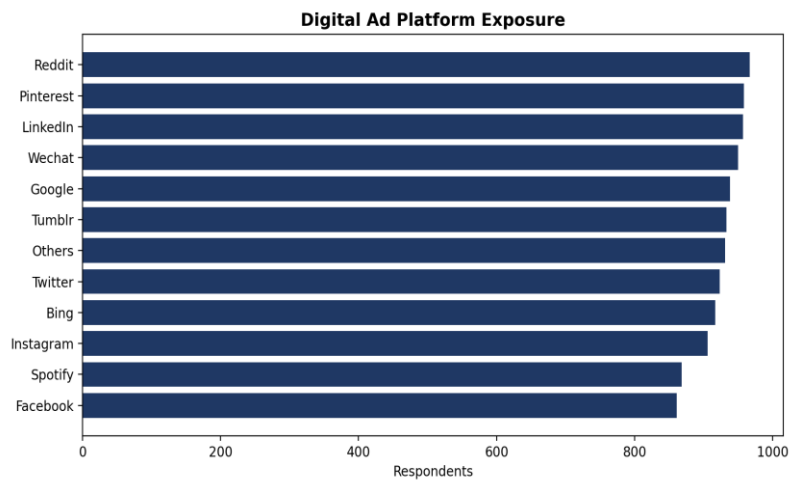


Figure 3: Respondents by digital ad platform exposure

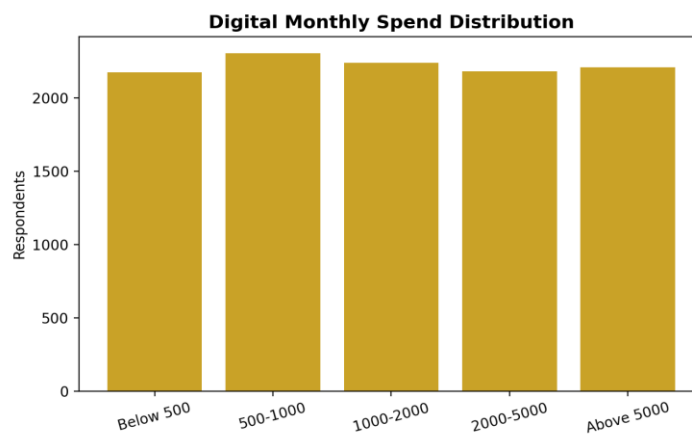


Figure 4: Digital monthly spend distribution

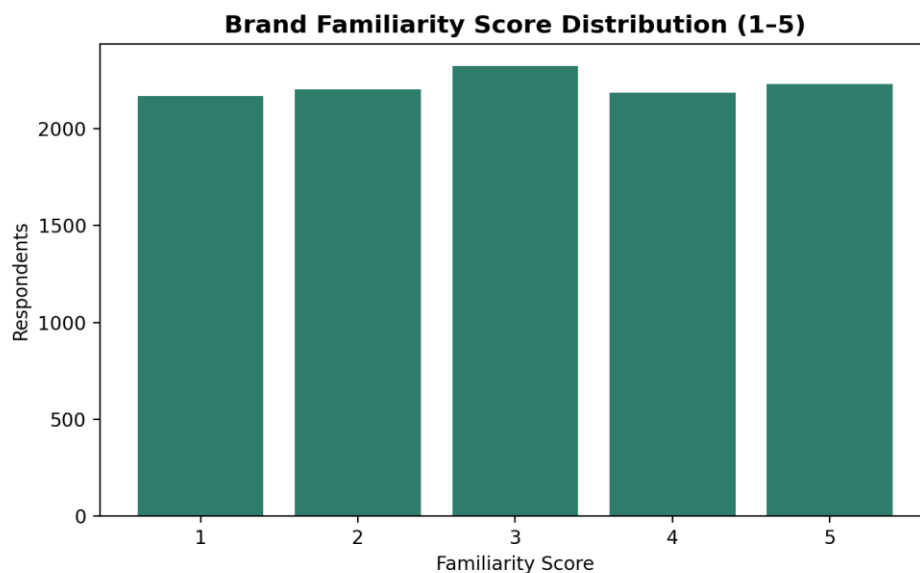


Figure 5: Brand familiarity score distribution (1–5)

6. Reliability Check — Test–Retest on Awareness

The questionnaire asked the awareness question twice (outlook_awareness and outlook_awareness_repeat). Comparing the two responses tests whether respondents answer consistently — a basic data-quality control before any driver analysis is trusted.

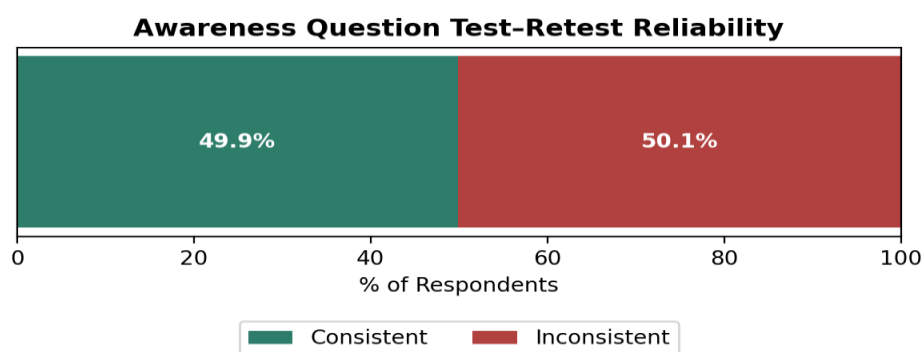


Figure 6: Test–retest agreement rate on the brand-awareness question

Metric	Value
Agreement rate	49.86%
Expected if random (2 binary options)	50.00%
Verdict	Statistically indistinguishable from chance

An agreement rate of 49.9% — effectively identical to the 50% expected from random guessing on a binary question — indicates the awareness measure has very poor test–retest reliability in this dataset. This is the single most important data-quality finding in this report and materially limits confidence in any awareness-driver conclusion drawn below.

7. Chi-Square Tests — Awareness vs Marketing Variables

Chi-square tests of independence checked whether each categorical marketing variable is associated with brand awareness (Yes/No).

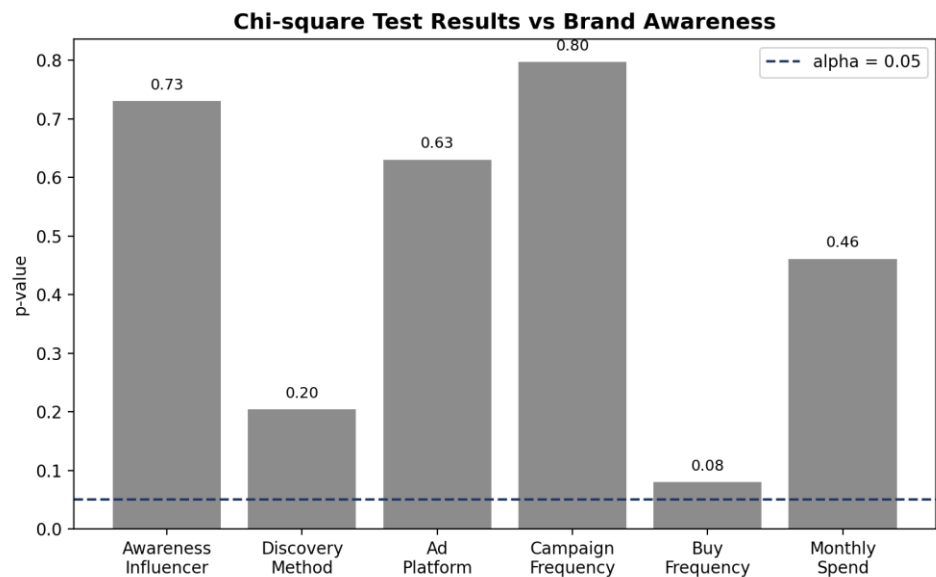


Figure 7: Chi-square p-values for awareness vs. each marketing variable (red = below 0.05 threshold)

Variable	Chi-square (χ^2)	df	p-value	Significant?
Brand Awareness Influencer	1.295	3	0.730	No
Brand Discovery Method	5.937	4	0.204	No
Digital Ad Platform	8.910	11	0.630	No
Campaign Frequency	1.019	3	0.797	No
Digital Buy Frequency	8.338	4	0.080	Borderline (not significant at 5%)
Digital Monthly Spend	3.611	4	0.461	No

None of the six marketing/exposure variables reach conventional significance ($p < 0.05$) against brand awareness. Digital buy frequency is the closest to significance ($p = 0.08$) and is worth monitoring in a follow-up wave, but on this data it does not meet the standard threshold.

8. Familiarity Score by Awareness — t-Test

Group	Mean Familiarity (1-5)	N
Aware (Yes)	3.006	5,486
Not Aware (No)	3.014	5,623

Statistic	Value
t-statistic	-0.295
p-value	0.768
Conclusion	No significant difference in familiarity between aware and unaware respondents

Counter-intuitively, respondents who say they are unaware of the brand rate their "familiarity" no lower than those who say they are aware — reinforcing the reliability concern raised in Section 6, since these two measures should logically move together.

9. Logistic Regression — Predicting Brand Awareness

A logistic regression modelled the probability of brand awareness (Yes = 1) as a function of brand-awareness influencer, digital monthly spend, campaign frequency, and familiarity score.

Model Statistic	Value
Pseudo R ²	0.0004
Log-Likelihood Ratio p-value	0.873
Significant predictors (p < 0.05)	None

The model's pseudo-R² of 0.0004 confirms the predictors jointly explain almost none of the variation in awareness, and the likelihood-ratio test (p = 0.873) shows the model is not a meaningful improvement over an intercept-only baseline. Individually, no coefficient is statistically significant, with digital_monthly_spend "Below 500" the closest at p = 0.075.

10. Pivot Table Cross-Tabulations

To complement the chi-square screening in Section 7, three additional pivot-style cross-tabulations were built to surface potential patterns hiding within specific variable pairs — the same approach applied to the workbook's pre-built pivots in Case Study 1.

10.1 PT1 — Brand Awareness × Digital Buy Frequency

This pairing was carried forward from Section 7 because it produced the closest-to-significant result ($p = 0.080$) of any variable tested against awareness, and merits a full pivot view.

Awareness	Weekly	Monthly	Regularly	Frequently	Yearly	Row Total
No	1,075	1,109	1,120	1,165	1,154	5,623
Yes	1,145	1,094	1,113	1,069	1,065	5,486
Column Total	2,220	2,203	2,233	2,234	2,219	11,109

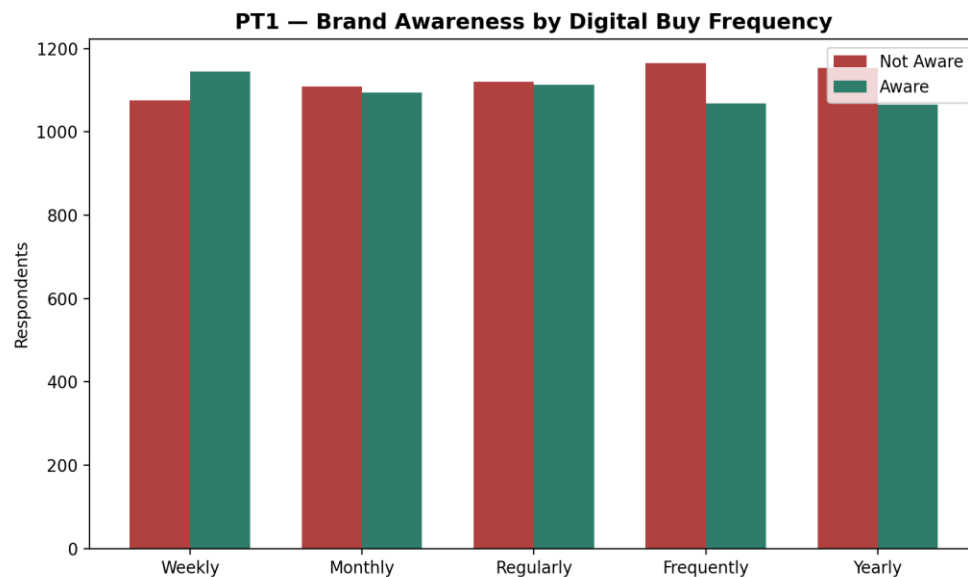


Figure 8: Brand awareness by digital buy-frequency segment

Statistic	Value
Chi-square (χ^2)	8.338
Degrees of Freedom	4
p-value	0.080
Conclusion	Not significant at 5%, but the closest signal in this study

"Frequently" and "Yearly" digital buyers skew slightly toward unawareness (1,165 and 1,154 "No" vs 1,069 and 1,065 "Yes"), while "Weekly" buyers skew slightly toward awareness (1,145 "Yes" vs 1,075 "No"). The pattern falls just short of the 5% significance threshold, but its direction — frequent digital shoppers are not necessarily more brand-aware — is worth tracking in a cleaner follow-up wave.

10.2 PT2 — Digital Monthly Spend × Campaign Frequency

This cross-tabulation checks whether customers exposed to more frequent digital campaigns also report higher digital spend — a basic test of marketing-to-revenue linkage.

Monthly Spend	1–5 Camp./Mo	6–10 Camp./Mo	11–20 Camp./Mo	20+ Camp./Mo	Row Total
Below 500	551	523	524	578	2,176
500–1000	627	573	557	547	2,304
1000–2000	536	569	581	553	2,239
2000–5000	541	553	527	561	2,182
Above 5000	544	535	592	537	2,208

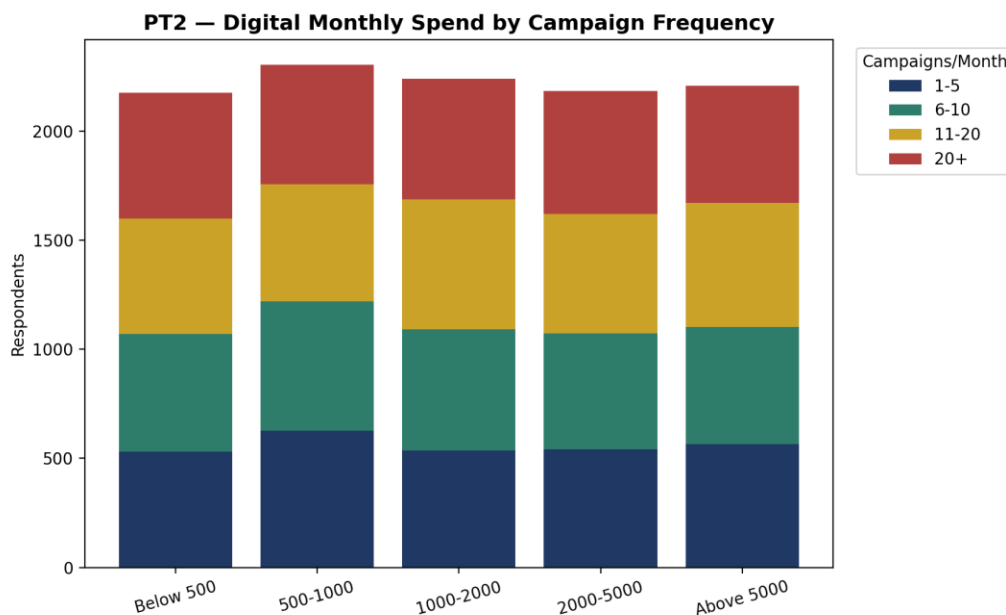


Figure 9: Digital monthly spend by campaign-frequency exposure

Statistic	Value
Chi-square (χ^2)	17.353
Degrees of Freedom	12
p-value	0.137
Conclusion	No significant association ($p > 0.05$)

Spend bands are near-identical in their campaign-frequency mix — customers exposed to 20+ campaigns a month are no more likely to be high spenders than those exposed to just 1–5. On this data, campaign volume alone does not appear to translate into higher digital spend.

10.3 PT3 — Brand Discovery Method × Awareness

This view checks whether the channel through which a customer first discovered the brand relates to their current stated awareness.

Discovery Method	Aware (Yes)	Not Aware (No)	Row Total
I myself read it	1,165	1,121	2,286
Others	1,055	1,121	2,176
People who I know read it	1,070	1,173	2,243
Print Media	1,090	1,092	2,182
Social Media	1,106	1,116	2,222

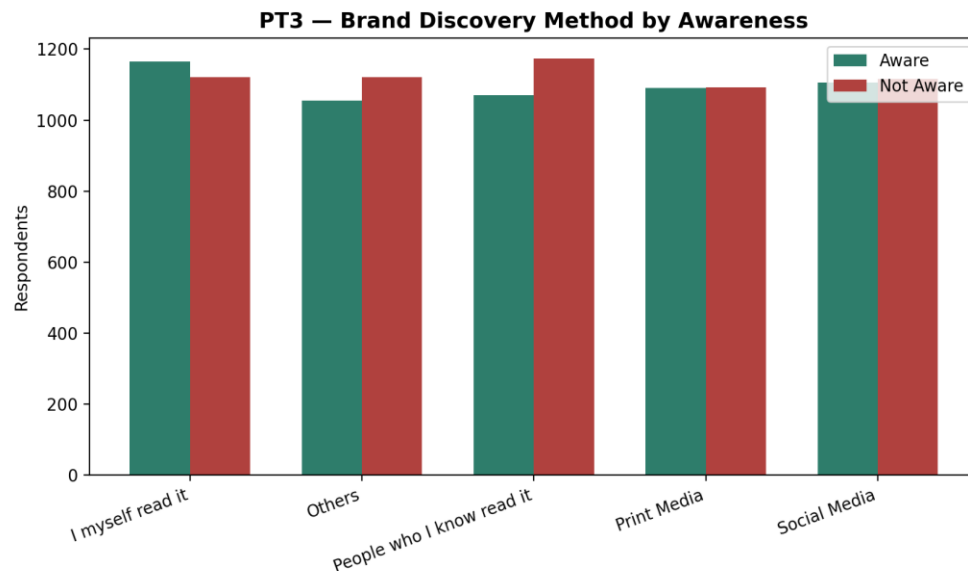


Figure 10: Brand discovery method by current awareness status

Statistic	Value
Chi-square (χ^2)	5.937
Degrees of Freedom	4
p-value	0.204
Conclusion	No significant association ($p > 0.05$)

Word-of-mouth discovery ("People who I know read it") shows the widest gap in this table — 1,173 unaware vs 1,070 aware — but the difference is not statistically significant at the 5% level. No discovery channel stands out as a reliably stronger awareness driver than another.

11. Interpretation & Business Implications

10.1 What the data tells us

- Brand awareness splits almost exactly 50/50, with no channel, platform, campaign band, or spend level showing a statistically significant association.
- The awareness question itself fails a basic test–retest reliability check (49.9% agreement, no better than chance).
- Familiarity score does not differ between aware and unaware respondents — a logical inconsistency that further supports a measurement problem rather than a true absence of drivers.

10.2 Likely explanations

- Survey-instrument reliability: an awareness question that respondents cannot answer consistently within the same survey cannot be used to infer real-world marketing effectiveness.
- Possible causes: ambiguous question wording, order/context effects between the two awareness questions, low respondent engagement, or synthetic/randomised response generation in this dataset.
- Because the reliability failure precedes the driver analysis, the chi-square and regression "non-findings" in Sections 7–9 cannot be interpreted as evidence that marketing channels don't matter — they may simply reflect noise in the outcome variable.

10.3 Recommendations

- Fix and re-field the awareness question before drawing any channel-investment conclusions — test single-question wording, remove duplicate placement, and pilot for consistency (target $\geq 85\%$ test–retest agreement).
- Once a reliable awareness measure exists, re-run the chi-square and logistic regression analysis in this report unchanged — the method is sound, only the input measure needs repair.
- In parallel, use digital_buy_frequency ($p = 0.08$, the closest signal found) as a starting hypothesis for a more targeted follow-up study.
- Audit overall survey data collection quality, since both Case Study 1 and Case Study 2 show uniform, low-signal response patterns across unrelated variable sets — a pattern worth escalating to the research vendor.

12. Conclusion

Case Study 2 applied the same rigorous Data Analysis pipeline — cleaning, descriptive profiling, reliability testing, chi-square, t-test and logistic regression — to eight variables representing brand awareness and digital marketing exposure. As with Case Study 1, no statistically significant relationship was found between marketing exposure variables and the outcome measure. Critically, this study also uncovered a direct reliability failure in the awareness question itself (49.9% test–retest agreement), which is the most actionable finding in this report: it points management toward fixing survey instrumentation as a prerequisite for any future marketing-effectiveness analysis, rather than concluding that channel or spend decisions are irrelevant to awareness.

BRAND AWARENESS & DIGITAL MARKETING — DASHBOARD

Case Study 2 | Outlook Brand Consumer Survey | n = 11,109 respondents

49.4%
BRAND AWARENESS
Say "Yes"

49.9%
TEST-RETEST
≈ random chance

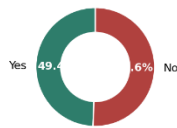
p = 0.08
BEST CHI-SQ SIGNAL
Buy frequency

3.01 / 5
FAMILIARITY SCORE
Mean, all resp.

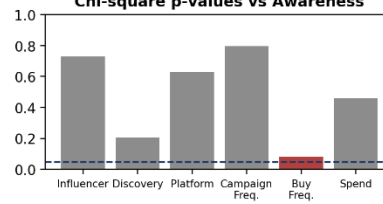
0 of 6
SIG. DRIVERS FOUND
No p<0.05 result

11,109
SAMPLE SIZE
Clean, 0% missing

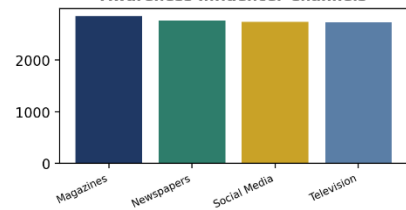
Brand Awareness Split



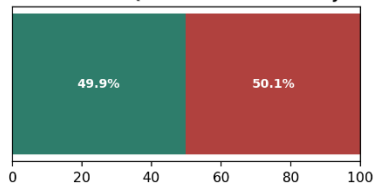
Chi-square p-values vs Awareness



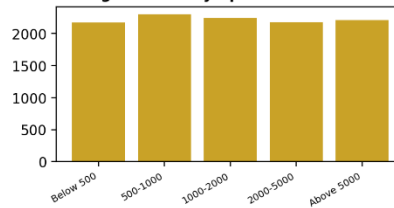
Awareness Influencer Channels



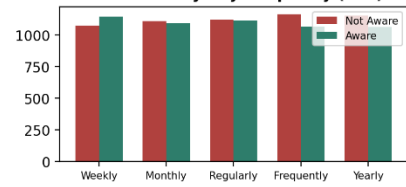
Awareness Q. Test-Retest Reliability



Digital Monthly Spend Distribution



Awareness by Buy Frequency (PT1)



Consistent Inconsistent

Awareness question fails test-retest reliability (49.9% agreement) — treat driver findings as provisional | Source: Outlook Brand Consumer Survey, n=11,109

THEORETICAL ANALYSIS

1. What is Data Visualization?

Answer:

Data Visualization is the graphical representation of data using charts, graphs, maps, dashboards, and other visual elements. It helps users understand complex data by presenting it in a simple and meaningful way. Data visualization enables businesses and individuals to identify trends, patterns, relationships, and outliers, making data-driven decision-making easier.

Importance of Data Visualization:

- Simplifies large and complex datasets.
- Helps identify trends and patterns quickly.
- Supports faster and better decision-making.
- Improves communication of insights.
- Detects anomalies and business opportunities.

2. What are the different techniques involved in Data Visualization?

Answer:

Various data visualization techniques are used depending on the type of data and the objective of analysis.

Technique	Purpose
Bar Chart	Compares values across different categories.
Line Chart	Shows trends over time.
Pie Chart	Displays percentage or proportion of a whole.
Histogram	Shows frequency distribution of continuous data.
Scatter Plot	Displays relationships between two variables.
Area Chart	Illustrates cumulative totals over time.
Heat Map	Represents data intensity using different colors.
Tree Map	Shows hierarchical data using nested rectangles.
Bubble Chart	Compares three variables using bubble size.
Geographical Maps	Visualizes location-based data.
Dashboard	Combines multiple visualizations into a single interactive view.
Pivot Charts	Creates dynamic visual summaries from Pivot Tables.

3. Which are the different software that helps in Data Visualization?

Answer:

Several software tools are available for creating visualizations and business reports.

Software	Description
Microsoft Excel	Creates charts, Pivot Tables, dashboards, and basic analytics.
Tableau	Popular Business Intelligence (BI) tool with interactive dashboards.
Microsoft Power BI	Creates real-time dashboards and business reports.
Google Looker Studio	Cloud-based reporting and dashboard tool.
Qlik Sense	Interactive analytics and data exploration platform.
Python (Matplotlib, Seaborn, Plotly)	Used for advanced visualization and data analysis.
R (ggplot2, Shiny)	Statistical analysis and high-quality graphics.
Apache Superset	Open-source business intelligence and dashboard software.
SAS Visual Analytics	Enterprise-level analytics and reporting solution.
IBM Cognos Analytics	Business reporting and visualization software.

4. What is Big Data? What are its characteristics?

Answer:

Big Data refers to extremely large and complex datasets that cannot be processed efficiently using traditional database systems. Big Data is generated from various sources such as social media, business transactions, sensors, IoT devices, online shopping, healthcare, and financial systems.

Characteristics of Big Data (5 Vs)

1. Volume - Refers to the enormous amount of data generated every second.
 2. Velocity - Indicates the speed at which data is generated, processed, and analyzed.
 3. Variety - Data exists in multiple formats such as structured, semi-structured, and unstructured.
 4. Veracity - Represents the accuracy, reliability, and quality of data.
 5. Value - Refers to the useful insights and business benefits obtained from analyzing data.
-

5. Which all Data Visualization techniques and software are you aware of?

Answer:

Data Visualization Techniques

- Bar Chart
- Line Chart
- Pie Chart
- Histogram
- Scatter Plot
- Area Chart
- Heat Map
- Tree Map
- Bubble Chart
- Box Plot
- Geographical Map
- Dashboard
- Pivot Charts

Data Visualization Software

- Microsoft Excel
- Microsoft Power BI
- Tableau
- Google Looker Studio
- Qlik Sense
- Python (Matplotlib, Seaborn, Plotly)
- R (ggplot2)
- Apache Superset
- IBM Cognos Analytics
- SAS Visual Analytics

These tools help organizations analyze data efficiently and present meaningful insights through interactive visual reports.

6. What is a Pivot Table?

Answer:

A Pivot Table is a data summarization tool available in Microsoft Excel and other spreadsheet software. It allows users to organize, analyze, summarize, and present large datasets without writing complex formulas.

Features of Pivot Tables

- Summarizes large amounts of data quickly.
- Groups and categorizes information.
- Performs calculations such as Sum, Average, Count, Maximum, and Minimum.
- Filters and sorts data easily.
- Generates Pivot Charts for visualization.
- Supports interactive business reporting.

Applications

- Sales analysis
 - Financial reporting
 - Inventory management
 - Customer segmentation
 - Performance analysis
-

7. How do dashboards aid Data Visualization?

Answer:

A dashboard is an interactive visual interface that combines multiple charts, graphs, KPIs, and reports into a single screen. It provides a comprehensive view of business performance and helps users monitor key metrics in real time.

Benefits of Dashboards

- Present multiple visualizations in one place.
- Monitor Key Performance Indicators (KPIs).
- Support real-time business decision-making.
- Identify trends, patterns, and anomalies quickly.
- Improve communication between departments.
- Allow users to filter and interact with data dynamically.
- Save time by eliminating manual report preparation.

Examples of Dashboard Applications

- Sales Dashboard
- Marketing Dashboard
- Financial Dashboard
- HR Dashboard
- Supply Chain Dashboard
- Customer Analytics Dashboard

Conclusion:

Dashboards transform raw data into meaningful visual insights, enabling managers and organizations to make faster, informed, and data-driven decisions.